6.1.1 Electromagnetism

	Learning outcomes	Additional guidance
(a)	*Describe and explain:*	
	(i) the action of a transformer: magnetic flux from a coil; induced e.m.f = rate of change of flux linkage	HSW2
	(ii) the action of a dynamo: change of flux linked produced by relative motion of flux and conductor	
	(iii) electromagnetic forces; qualitatively as arising from tendency of flux lines to contract or interaction of induced poles; quantitative calculation limited to force on a straight current-carrying wire in a uniform field	
	(iv) simple linked electric and magnetic circuits: flux produced by current turns, need for large conductance and permeance and the effect of increasing the dimensions of an electromagnetic machine; qualitative effect of iron and air gap.	permeance is understood as a magnetic equivalent to electrical conductance and with analogous dependence on the dimensions and nature of the magnetic medium
(b)	*Make appropriate use of:*	
	(i) the terms: **B**-field, magnetic field, flux, flux linkage, induced e.m.f, eddy currents	
	by sketching and interpreting:	
	(ii) graphs of variations of currents, flux and induced e.m.f	
	(iii) diagrams of lines of flux in magnetic circuits; continuity of lines of flux.	

(c) *Make calculations and estimates involving:*

(i) $\phi = BA$, $\mathcal{E} = -\dfrac{\mathrm{d}(\phi N)}{\mathrm{d}t}$	In the context of Faraday's and Lenz's laws HSW1, 2, 8
(ii) $F = ILB$	Calculations restricted to current perpendicular to uniform magnetic field
(iii) $\dfrac{V_1}{V_2} = \dfrac{N_1}{N_2}$ for an ideal transformer	i.e. no energy losses for an ideal transformer Learners will also be expected to recall the equation relating *V* and *N*
(iv) $\dfrac{I_2}{I_1} = \dfrac{N_1}{N_2}$ for an ideal transformer	Learners will also be expected to recall the equation relating *I* and *N* HSW12

(d) *Demonstrate and apply knowledge and understanding of the following practical activities (HSW4):*

(i) observing induced e.m.fs produced under varying conditions such as dropping a magnet through a coil attached to a data logger or oscilloscope	links to **6.1.1a(ii), 6.1.1c(i)**
(ii) determining the uniform magnetic flux density between the poles of a magnet using a rigid current carrier and digital balance.	links to **6.1.1a(iii), c(ii)**
(iii) investigate transformers.	links to **6.1.1a(i), c(iii)(iv)**